

ORIGINAL ARTICLE

Efficacy of mentalization-based therapy in treating self-harm: A systematic review and meta-analysis

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Funding information

Austrian Science Fund (FWF), Grant/Award Number: KLI963-B; Medical University of Vienna

Abstract

Introduction: Mentalization-based therapy (MBT) and its adapted version for adolescents (MBT-A) are repeatedly highlighted as promising treatments for reducing self-harm, particularly in borderline personality disorder (BPD). Despite the availability of publications providing evidence of their efficacy in reducing self-harm, recent meta-analyses have yielded mixed results. To inform best-practice clinical decision-making, we conducted a systematic review and meta-analysis. We aimed to disentangle findings for both adolescents and adults on the efficacy of MBT(-A) in reducing self-harm (primary outcome) and symptoms of BPD and depression (secondary outcomes).

Methods: Web of Science, Scopus, Embase, PubMed/Medline, and Cochrane Review Database were searched for eligible studies published until September 2022. In total, 14 studies were identified, comprising 612 participants from nine MBT studies (six pre-post, three RCTs) and five MBT-A studies (two pre-post, three RCTs). Aggregated effect sizes were estimated using random-effects models. Meta-regressions were conducted to assess the effect of moderator variables (treatment duration, drop-out rates, and age) on effect sizes.

Results: Overall, both MBT and MBT-A demonstrated promising effects in reducing self-harm ($g = -0.82$, 95% CI -1.15 to -0.50), borderline personality disorder ($g = -1.08$, 95% CI -1.38 to -0.77), and depression ($g = -1.1$, 95% CI -1.52 to -0.68) symptoms. However, when compared to control interventions (TAU, SCM), MBT(-A) did not prove to be more efficacious, with the exception of MBT showing superior effects on BPD symptoms in adults ($g = -0.56$, 95% CI -0.88 to -0.24).

Conclusion: Although the pre-post evaluations seem promising, this analysis, including RCTs, showed no superiority of MBT(-A) to control conditions, so that prioritizing the application of MBT (-A) for the treatment of self-harm is not supported. Possible explanations and further implications are discussed.

Carola Hajek Gross and Sofia-Marie Oehlke shared first authorship.

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KEYWORDS

borderline personality disorder, depression, MBT, self-harm, self-injury

INTRODUCTION

Self-harm is a serious and widespread mental health problem that typically commences and peaks in adolescence (Gillies et al., 2018; Plener et al., 2015). Although prevalence rates are higher in adolescence (22.9%, Gillies et al., 2018), self-harm is also represented among adults (5.5%, Swannell et al., 2014; Troya et al., 2019). In this review, self-harm is defined according to Hawton et al. (2012) as intentional self-poisoning or self-injuring acts, regardless of the underlying motivation or degree of suicidal intent.

Critically, self-harm irrespective of suicidal intention is known to be a major risk factor for suicidal deaths (Castellví et al., 2017). Self-harm is strongly associated with psychopathology (Nock et al., 2006), and is especially frequent in patients with borderline personality disorder (BPD). Among inpatients with BPD, up to 95% of adolescents and 90% of adults report a history of self-harm, with 51%–65% having engaged in at least 25 episodes (Goodman et al., 2017). Given the clinical significance and high prevalence of self-harm in both adolescents and adults, establishing effective treatments is key. Recent meta-analytic evidence suggests overall small effect sizes of self-harm interventions (Fox et al., 2020; Kothgassner et al., 2020). However, there is limited evidence for small to moderate effect sizes for dialectical behavioral therapy for adolescents (DBT-A) in reducing self-harm (Kothgassner et al., 2020, 2021).

One promising evidence-based approach in the presence of self-harm is mentalization-based therapy (MBT) by Bateman and Fonagy (1999), which is a structured disorder-specific psychotherapy that has been recommended for BPD treatment (Lieb & Stoffers-Winterling, 2020; Stoffers-Winterling & Lieb, 2022). The main target of the treatment is to improve mentalizing—the ability to interpret the behavior of oneself and others in terms of underlying mental states (e.g., intentions, thoughts) (Fonagy et al., 2018). Difficulties in emotion regulation and interpersonal functioning, which are centrally implicated in BPD and self-harm (Reichl & Kaess, 2021), are theorized to arise when mentalizing capacity is impaired (Sharp et al., 2012; Sharp & Kalpakci, 2015). Fonagy and Bateman (2006) argue that mentalizing is an important prerequisite for therapeutic processes to be effective. In fact, a recent systematic review found the initial capacity of mentalizing predictive of treatment outcomes, although longitudinal evidence

of the role of mentalizing in the therapeutic process remains tentative (Lüdemann et al., 2021).

MBT is a manualized treatment that integrates aspects of psychoanalysis, attachment theory, and cognitive theory (Bateman & Fonagy, 1999). MBT was initially developed as a high-frequency partial hospitalization program on 5 days per week for 18 months, including mentalization-based individual and group therapy sessions additionally to other day hospital program (Bateman & Fonagy, 1999; Karterud, 2015). As treatment results were not clearly attributable to the mentalization-based treatment components, a more purified program was later designed for outpatient use (Bateman & Fonagy, 2006; Karterud, 2015). MBT in outpatient settings includes two individual and group therapy sessions per week for 18 months. MBT in a group format (MBT-G) has been designed as a 12–36 months program offering solely group and family sessions without individual sessions, representing a cost-effective alternative to multicomponent MBT-programs (Jørgensen et al., 2021; Karterud, 2015). While MBT was originally developed to treat adults with BPD, it has further been adapted for treating adolescents (MBT-A). MBT-A has a typical duration of 12 months and includes weekly or monthly family sessions in addition to or instead of group therapy sessions (Taubner et al., 2017; Taubner & Sevecke, 2015). As MBT, MBT-A can be applied in both partial hospitalization settings held twice per week and outpatient settings held once per week (Taubner & Sevecke, 2015).

Previous meta-analyses and systematic reviews in the scope of therapies for treating BPD yielded mixed evidence for mentalization-based treatment in reducing self-harm (Jørgensen et al., 2021; Malda-Castillo et al., 2019; Stoffers-Winterling et al., 2022; Storebø et al., 2020; Vogt & Norman, 2019). However, no current meta-analysis and systematic review focuses specifically on the application of MBT(-A) for treating self-harm. Additionally, there is some evidence that findings may diverge for different age-groups (Jørgensen et al., 2021; Stoffers-Winterling et al., 2022). Yet, no clear meta-analytic comparison between the two different approaches for adults (MBT) and adolescents (MBT-A) has been established. Following this, we aim to provide an overview by contrasting the application of mentalization-based concepts for both adults and adolescents with a primary focus on self-harm. In addition to RCTs, we have included pre–post evaluation studies to adopt a more comprehensive approach, illustrating both

the effectiveness and efficacy of MBT(-A) interventions. In addition, we investigate whether variations among studies (participant age, treatment duration, and dropout rates) moderate the meta-analytic effect of MBT(-A) on our outcome measures.

METHODS

Search strategy and inclusion criteria

This meta-analysis was preregistered with the Open Science Framework (<https://doi.org/10.17605/OSF.IO/26JVF>). We conducted a search of Web of Science, Scopus, Embase, PubMed/Medline, and Cochrane Review Database for studies published until 30 September 2022 using the keywords (“MBT*” OR “Mentalization-based” OR “Mentalisation-based”) AND (“self harm” OR “self-harm” OR “NSSI” OR “self-injury” OR “self injury”). On Google Scholar, an alert was set up with the keywords when the search began. Additional studies were searched by screening reference lists of studies included. Studies were eligible to be included in the meta-analysis if (1) they were classified as RCTs or pre-post evaluations of MBT or MBT-A protocols, (2) reported severity scores for self-harm at baseline and posttreatment, and (3) included clinical samples.

The frequency of self-harm episodes was assessed as the primary outcome measure, and BPD and depression symptom scores (see study characteristics) were assessed as secondary outcome measures. We screened the title, abstract, and main text of each study for eligibility, excluding documents at each stage. The initial search yielded 369 results (see Figure 1). Full-text articles were screened independently by two authors (CHG and SO) for the final evaluation of inclusion.

Data extraction and analysis

Outcome measures (means and standard deviations) and study characteristics of included studies were independently extracted and entered into a spreadsheet by two authors (CHG and SO). A third author (ODK) reviewed and discussed differences until consensus was reached. If variables were not reported as means or standard deviations, conversions via Revman Calculator (Cochrane Collaboration, 2014) or formulas (Card, 2012) were conducted. Self-harm episodes reported as proportions or odds ratios were transformed to Hedges' g via the formula provided in Lipsey and Wilson (2001). If outcome measures were not reported in studies or Supporting Information, the authors of the respective study were contacted.

The analyses were conducted using the metafor package for R (Viechtbauer, 2010). For analyses regarding pre-post treatment effects, we computed the standardized mean difference (Hedges' g) based on means and standard deviations before and after MBT(-A) intervention using the formula $g = (M_{\text{post}} - M_{\text{pre}}) / SD_{\text{pooled}}$, where M_{post} is the mean of the measure after the intervention and M_{pre} the mean before the intervention, with SD_{pooled} as the standard deviation for both measurements, defined as $SD_{\text{pooled}} = \text{SQRT}((SD_{\text{pre}}^2 + SD_{\text{post}}^2) / 2)$ (Lakens, 2013). We further conducted subgroup analyses regarding the pre-post treatment-effects of MBT-A and MBT studies. To assess the efficacy of MBT(-A) in reducing self-harm, BPD, and depression compared to control interventions, we computed the standardized mean difference (Hedges' g) of posttreatment scores based on means and standard deviations of MBT(-A) and control groups and further conducted subgroup analyses for MBT-A and MBT studies.

Random-effects models were applied to estimate aggregated effect sizes. Following established conventions (Cohen, 1988), an effect size of 0.20 was considered a small effect, 0.50 a medium effect, and 0.80 a large effect. Heterogeneity across study outcomes was reported with I^2 values, where 25% indicated low, 50% moderate, and 75% high heterogeneity (Higgins et al., 2003).

Moderator analyses (meta-regression) were conducted to assess whether treatment duration in months, dropout rates in the MBT(-A) group and mean age in years moderated the effect of MBT(-A) on each outcome (see Appendix S1). A sensitivity analysis was conducted excluding the pilot study by Griffiths et al. (2019), which was limited to 12 weeks of psychoeducation and experiential exercises. Subgroup analyses were further conducted, to differentiate between MBT-A treatment settings (individual vs. group). As most analyses included <10 studies (Higgins & Green, 2011), publication bias is not reported. Nevertheless, Egger's regressions were conducted to estimate publication bias via funnel plot asymmetry (Sterne & Egger, 2006), applying the trim-and-fill method to calculate effect sizes adjusted for missing studies (Duval & Tweedie, 2000) (see Appendix S2, Figures S1A–C and S2A–C).

Risk of bias assessment

The risk of bias for the primary outcome (self-harm) was independently assessed by three authors (KP, SO, and CHG) using the risk of bias (RoB 2) tool for randomized controlled trials (Sterne et al., 2019). Any disagreements were settled through discussions between the authors. The likelihood of bias was assessed in regard to the following

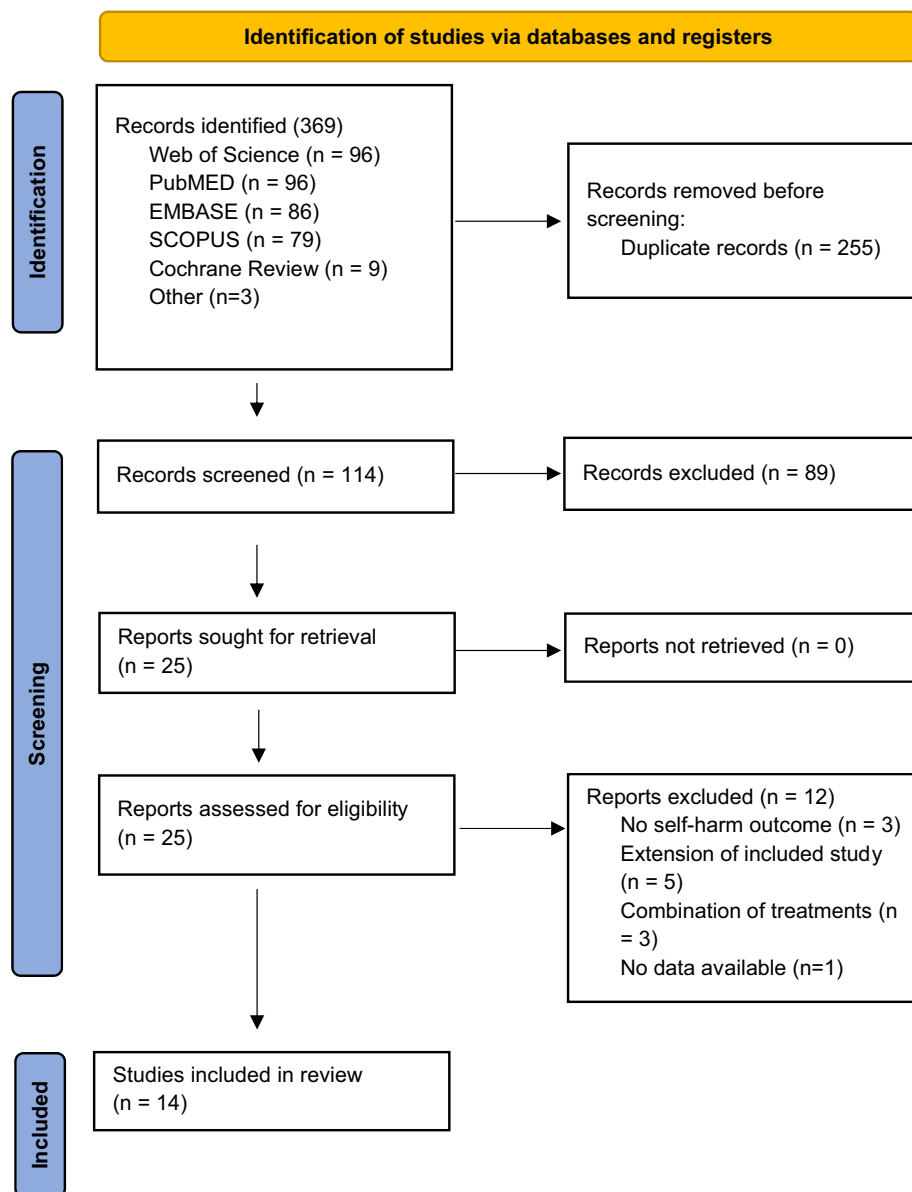


FIGURE 1 PRISMA flowchart detailing the study selection process.

domains: bias due to (a) the randomization process, (b) deviations from the intended interventions, (c) missing outcome data, (d) the measurement of the outcome, and (e) selection of the reported result. For each domain, “signaling questions” were answered with “yes,” “probably yes,” “probably no,” “no,” and “no information.” Each study was subsequently judged as “low risk of bias,” “some concerns,” or “high risk of bias” by an algorithm and by a rater. The rater assessment overwrote the algorithm assessment in case of disagreement. The judgments within each domain led to an overall risk-of-bias judgment. A “low risk of bias” signifies the study was judged to be at low risk of bias for all domains. “Some concerns” indicate that the study was judged to be at some concerns in at least one domain. A “high risk of bias” indicates the study was judged to be at high risk of bias in at least one

domain or to have some concerns for multiple domains in a way that substantially invalidated the results (Sterne et al., 2019).

RESULTS

Study characteristics

In total, $k = 14$ samples were included in the analysis (see Table 1), of which five covered adolescent samples (MBT-A: two pre-post trials, three RCTs) and nine covered adult samples (MBT: Six pre-post trials, three RCTs). Adding to previous literature (Jørgensen et al., 2021; Stoffers-Winterling et al., 2022; Storebø et al., 2020), we identified one further RCT (Griffiths et al., 2019).

Out of all studies covering adolescent samples, Griffiths et al. (2019) conducted a pilot RCT of only a brief mentalization-based introductory intervention in groups (MBT-Ai) in combination with TAU. Rossouw and Fonagy (2012) conducted MBT-A in an individual format only, while Bo et al. (2017), Beck et al. (2020), and Tromp et al. (2021) conducted MBT-G for adolescents in a group format only. To consider the influence of different treatment formats, we conducted separate sensitivity analyses excluding Griffiths et al. (2019) (see [Results: Exploratory sensitivity analyses](#)) as well as Rossouw and Fonagy (2012) (see [Appendix S3, Figures S1A–C and S2A–C](#)).

Out of all studies covering adult samples, the study by Smits et al. (2020) was treated as two separate pre-post evaluation studies, as they report measures for two types of MBT interventions—a day hospital (MBT-DH) and an intensive outpatient (MBT-IOP) treatment. The study by Barnicot and Crawford (2019)—originally a non-randomized comparison of clinical outcomes between patients receiving either DBT or MBT—was treated as a pre-post evaluation study of MBT outcomes only. The study by Kvarstein et al. (2015)—originally a naturalistic, longitudinal comparison of treatment effects between MBT and a former psychodynamic program—was also treated as a pre-post evaluation study of MBT outcomes only.

In the following, all studies covering adolescent samples will be referred to as MBT-A, irrespective of treatment format. Equally, all studies covering adult samples will be referred to as MBT.

Overall, the included studies covered 612 participants, of which 257 received MBT, 193 received MBT-A, and 162 received a control intervention. Control interventions were either Treatment-as-Usual (TAU) or Structural Clinical Management (SCM). Participants receiving MBT-A were, on average, 16.1 (SD=0.8) years old and predominantly female ($M=88\%$) and participants receiving MBT were, on average, 31.1 (SD=3.0) years old and predominantly female ($M=79\%$). Dropouts, on average, were 31.4% in MBT-A (range: 8%–56%) and 35.1% (range: 0%–71%) in MBT studies.

Pre-post comparisons of MBT(-A)

Regarding overall pre-post comparisons of outcomes following all MBT(-A) studies (combined adolescents/adults) (see [Figure 2](#)), significant effects were found on self-harm ($g=-0.82$, 95% CI -1.15 to -0.50 , $k=14$), BPD ($g=-1.08$, 95% CI -1.38 to -0.77 , $k=12$), and depression ($g=-1.1$, 95% CI -1.52 to -0.68 , $k=7$) scores. Overall, participants showed large within-subject changes following MBT(-A) intervention across all outcomes. Heterogeneity between

studies was high regarding self-harm ($I^2=78.6\%$) and moderate-to-high regarding BPD ($I^2=59.9\%$) and depression ($I^2=62.9\%$).

Subgroup analyses of MBT-A studies (adolescents only) showed significant within-subject changes across all outcome scores, revealing moderate effect sizes on self-harm ($g=-0.56$, 95% CI -1.05 to -0.07 , $k=5$) and large effect sizes on BPD ($g=-0.90$, 95% CI -1.37 to -0.42 , $k=4$) and depression ($g=-0.81$, 95% CI -1.32 to -0.30 , $k=4$). Heterogeneity between MBT-A studies was high regarding self-harm ($I^2=74.4\%$) and moderate regarding BPD ($I^2=45.3\%$) and depression ($I^2=55.5\%$).

Subgroup analyses of MBT studies (adults only) also showed significant within-subject changes across all outcome scores, representing large effect sizes on self-harm ($g=-0.96$, 95% CI -1.37 to -0.56 , $k=9$), BPD ($g=-1.16$, 95% CI -1.55 to -0.76 , $k=8$), and depression ($g=-1.45$, 95% CI -1.93 to -0.96 , $k=3$). Heterogeneity between MBT studies was high regarding self-harm ($I^2=74.4\%$), moderate-to-high regarding BPD ($I^2=65.8\%$), and low-to-moderate regarding depression ($I^2=36.7\%$).

Comparisons of MBT(-A) to control interventions

Regarding the overall efficacy of MBT(-A) (combined adolescents/adults) compared to control interventions in reducing self-harm (see [Figure 3a](#)), we computed six effect sizes covering 312 participants (MBT-A: $n=66$, MBT: $n=84$, Control: $n=162$, $k=6$). The meta-analysis yielded no significant difference in self-harm scores between groups. Heterogeneity between studies was moderate-to-high ($I^2=68.2\%$). Subgroup analyses of MBT-A (adolescents only) and MBT studies (adults only) also found no significant differences between groups. Heterogeneity was low among both MBT-A ($I^2=29.6\%$) and MBT ($I^2=29.6\%$) studies. Overall, MBT(-A) was not superior to control interventions in reducing self-harm scores.

Regarding the overall efficacy of MBT(-A) (combined adolescents/adults) compared to control interventions in improving symptoms of BPD (see [Figure 3b](#)), we computed six effect sizes covering 312 participants (MBT-A: $n=66$, MBT: $n=84$, Control: $n=162$, $k=6$). The meta-analysis yielded no significant difference in BPD scores between groups across all MBT(-A) studies. Heterogeneity between studies was moderate ($I^2=48.0\%$). There were also no significant differences between groups in subgroup analyses of MBT-A studies (adolescents only). However, subgroup analyses of MBT studies (adults only) yielded a significant difference in BPD scores between groups ($g=-0.56$, 95% CI -0.88 to -0.24), representing moderate effect sizes. Heterogeneity was very low among both

TABLE 1 Study characteristics.

Study	Outcome (measures)	Mean age		% Female		Patients with BPD	Therapeutic intervention		
		MBT (-A)	Control	MBT (-A)	Control	<i>n</i> (%)	Subgroup	Format	Setting
Griffiths et al. (2019)	SH (RTSHI-A), BPD (BPFS-C), D (RCADS)	15 years	16 years	77%	81%	16 (33%)	<i>MBT-A</i>	Group	CAMHS (Child and Adolescent Mental Health Services)
Beck et al. (2020)	SH (RTSHI-A), BPD (BPFS-C), D (BDI-Y)	16 years	16 years	100%	98%	66 (100%)	<i>MBT-A</i>	Group	Outpatient
Rossouw and Fonagy (2012)	SH (RTSHI), BPD (BPFS-C), D (MFQ)	15 years	15 years	85%		27 (73%)	<i>MBT-A</i>	Individual	Outpatient
Bo et al. (2017)	SH (RTSHI-A), BPD (BPFS-C), D (BDI-Y)	16 years		100%		25 (100%)	<i>MBT-A</i>	Group	Outpatient

Treatment	n	Control interventions		Duration Means	Dropout	
		Treatment	n		MBT (-A)	Control
MBT-Ai plus TAU a. 12 sessions (1.25 h) per group Adaptions to MBT manual: experiential exercises and more focus on applying MBT skills in daily life, including psychoeducation	22	TAU Any combination of a. key worker input b. psychological/psychosocial intervention c. medication 26% of the TAU-alone group received a 7-week Distress Tolerance group intervention, based on the principles of DBT; 19% of participants were referred from Tier IV services, whose staff had previously been trained in AMBIT	26	12 weeks	2	1
MBT-G b. MBT-Introduction (3 sessions) c. MBT-Groups (37 weekly sessions, 90 min) d. MBT-Parents (6 sessions)	24	TAU a. at least 12 individual supportive sessions (one per month, but was allowed to vary) b. Psychoeducation c. Counseling d. If needed, ad hoc crisis management and sessions with caregiver participation Therapists were carefully selected to ensure they were not trained in or practicing MBT	42	12 months	31	14
MBT-A a. Weekly individual MBT-A sessions (50 mins) b. Monthly mentalization-based family therapy (50 mins)	20	TAU a. Evidence- based interventions according to diagnostic criteria Most received individual therapeutic intervention alone (28%), consisting of counseling (in 38% of cases receiving an individual intervention), generic supportive interventions (24%), cognitive behavioral therapy (19%), or psychodynamic psychotherapy (19%); a combination of individual therapy and family work (25%); or psychiatric review alone (27.5%)	17	12 months	20	23
MBT-G a. 34 sessions of MBT group-therapy (1.5 h) b. two individual case-formulation sessions (1 h each session) c. six group-based MBT-I (introductory sessions (1.5 h) d. MBT-Parents (7 sessions)	25			8.3 months	9	

(Continues)

TABLE 1 (Continued)

Study	Outcome (measures)	Mean age		% Female		Patients with BPD <i>n</i> (%)	Therapeutic intervention		
		MBT (-A)	Control	MBT (-A)	Control		Subgroup	Format	Setting
Tromp et al. (2021)	SH (DAPP-SF-A)	18 years		76%		BPD: n.a. Personality Pathology: 102 (100%)	MBT-A	Group	Inpatient
Philips et al. (2018)	SH (DSHI-9), BPD (BPDSI-IV)			80%		24 (100%)	MBT	Individual and group	Outpatient addiction treatment services
Bateman and Fonagy (1999)	SH (number of people who stopped self-harm), BPD (GSI), D (BDI)	30 years	33 years	68%	47%	38 (100%)		Individual and group	Day hospital
Bateman and Fonagy (2009)	SH (average count of severe sh incidents), BPD (GSI), and D (BDI)	31 years	31 years	80%	79%	99 (100%)	MBT	Individual and group	Outpatient
Löf et al. (2018)	SH (DSHI-9), BPD (KABOSS-S)	30 years		89%		22 (100%)	MBT	Individual and group	Outpatient

Treatment	<i>n</i>	Control interventions		Duration Means	Dropout	
		Treatment	<i>n</i>		MBT (-A)	Control
MBT-A MBT treatment (5 days a week) a. sociotherapy (continuously) b. group psychotherapy (twice a week) c. psychodrama (once a week) d. psychomotor therapy (once a week) e. creative therapy (once a week) f. family therapy (MBT-F; once a week) g. MBT-oriented group therapies (22 h a week) h. Parent group (every 6 weeks, 4 days/week) Individual psychotherapy was offered to approximately 5% of the study population, and EMDR to 5%–10%	102			12 months	19	
MBT (+SUD) a. individual therapy b. group therapy c. standard substance use disorder treatment (SUD)	13	SUD (standard substance use disorder treatment)	11	18 months	11	11
MBT a. individual psychotherapy (once a week) b. group psychotherapy (1 h, thrice a week) c. expressive therapy oriented toward psychodrama techniques (1 h, once a week) d. community meeting (1 h, once a week) e. meeting with the case administrator and medication review (1 h, once a month)	19	SCM (structured clinical management) a. regular psychiatric review with a senior psychiatrist when necessary (on average, twice per month) b. inpatient admission as appropriate with discharge to nonpsychoanalytic psychiatric partial hospitalization focusing on problem solving c. outpatient and community follow-up d. every-2-week visits (by a community psychiatric nurse) as standard aftercare No formal psychotherapy.	19	18 months	6	0
MBT a. weekly individual psychotherapy b. weekly group therapy	52	SCM (structured clinical management) a. regular individual sessions b. regular group sessions c. appointments every 3 months for psychiatric review Therapy was based on a counseling model closest to a supportive approach with case management, advocacy support, and problem-oriented psychotherapeutic interventions	47	18 months	19	16
MBT a. Introductory psycho-educational component (9–12 sessions) b. individual session (once a week) c. MBT group sessions (twice a week) d. expressive group sessions (twice a week)	22			18 months	53	

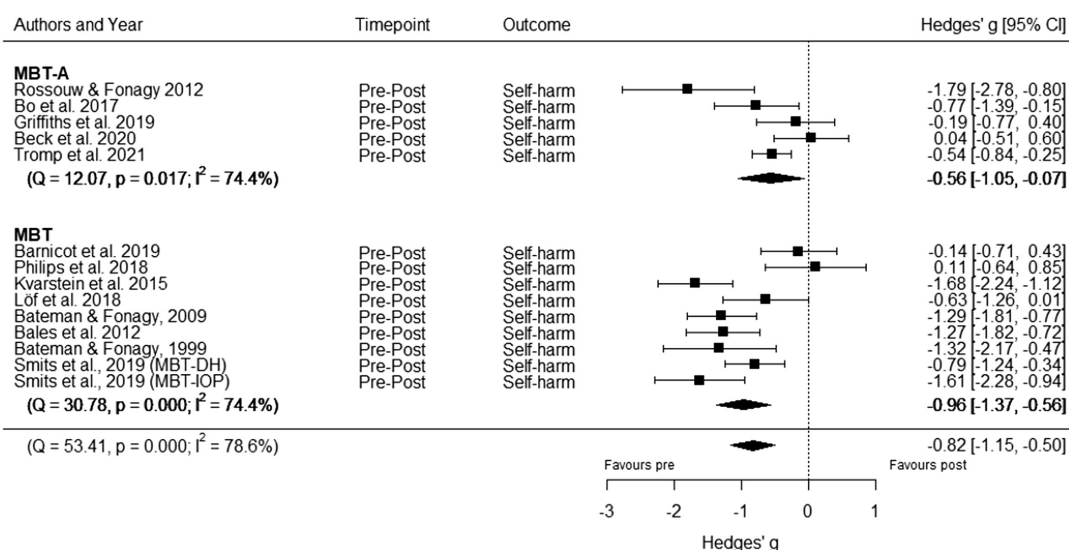
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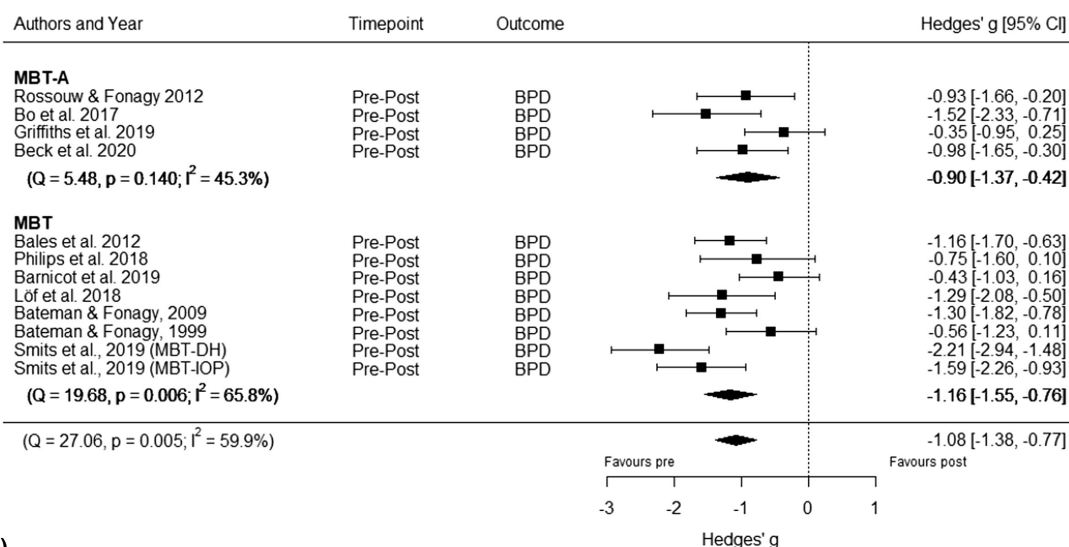
Study	Outcome (measures)	Mean age		% Female		Patients with BPD <i>n</i> (%)	Therapeutic intervention		
		MBT (-A)	Control	MBT (-A)	Control		Subgroup	Format	Setting
Bales et al. (2012)	SH (SSHI), BPD (BPDSI), and D (BDI)	30 years		71%		45 (100%)	MBT	Individual and group	Day hospital
Barnicot and Crawford (2019)	SH (SASII), BPD (BEST), and ER (DERS)	34 years		72%		23 (100%)	MBT	Individual and group	Outpatient
Kvarstein et al. (2015)	SH (incidents of self-harming)	26 years		84%		44 (100%)	MBT	Individual and group	Day hospital
Smits et al. (2020) Group A	SH (SSHI), BPD (PAI-BOR)	31 years		84%		50 (100%)	MBT	Individual and group	Day hospital
Smits et al. (2020) Group B	SH (SSHI), BPD (PAI-BOR)	30 years		80%		39 (100%)	MBT	Individual and group	Outpatient

Treatment	n	Control interventions		Duration	Dropout	
		Treatment	n	Means	MBT (-A)	Control
MBT-DH 5 days per week and 4.5 h per day a. implicit mentalizing groups (i.e., daily group psychotherapy and weekly individual psychotherapy, and individual crisis planning from a mentalizing perspective) b. explicit mentalizing group (i.e., art therapy twice a week, mentalizing cognitive group therapy, and writing therapy)	45			18 months	0	
MBT a. individual therapy (weekly or fortnightly) b. group therapy (weekly) c. Short-term group programme for psychoeducation and support (weekly, delivered over a ten-week period)	23	(vs. DBT, not included)		12 months	32	
MBT a. First year: weekly sessions of individual MBT, 12 sessions in an MBT psychoeducational group, and once weekly (1.5 hr) MBT dynamic group b. Second year: individual therapy sessions every second week c. Third year: frequencies of individual therapy sessions were reduced to every third week Weekly MBT group sessions continued throughout treatment	44	(vs. psychodynamic treatment program, not included)		36 months	43.52	
MBT-DH 5 days per week, nine group therapy sessions per week a. group psychotherapy (five times a week) b. group art therapy/writing (four times a week) c. therapy/mentalizing cognitive therapy d. social hour and community meeting e. weekly individual psychotherapy f. individual crisis management (on average weekly for 3 months, gradually lowering in frequency) g. psychiatric consultation upon request h. Pretreatment including a 12-session psychoeducation group	50			18 months	9	
MBT-IOP two group therapy sessions per week a. weekly individual psychotherapy b. individual crisis management (on average weekly for 3 months, gradually lowering in frequency) c. psychiatric consultation upon request d. Pretreatment including a 12-session psychoeducation group	39			18 months	5	

(a)



(b)



(c)

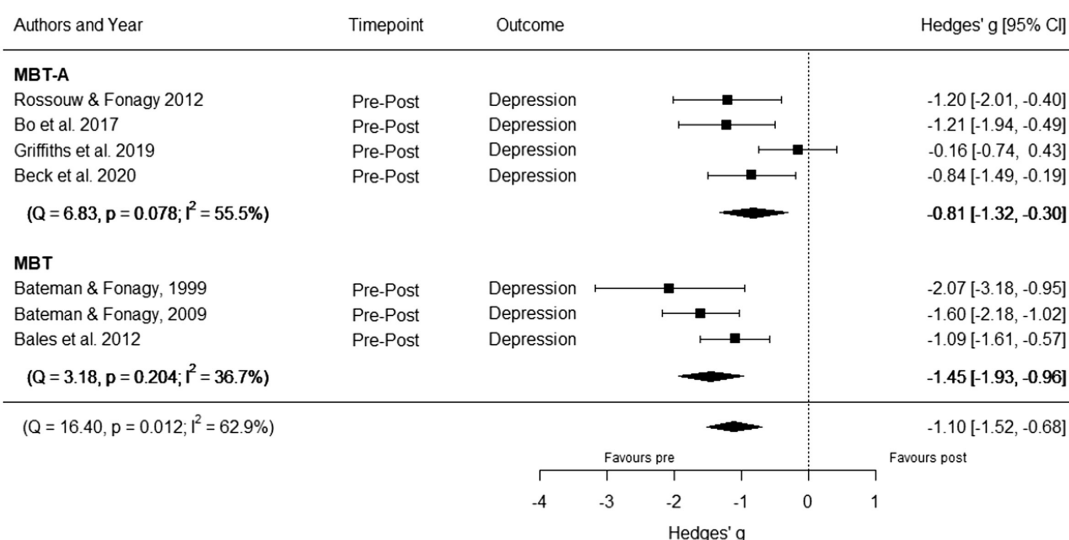
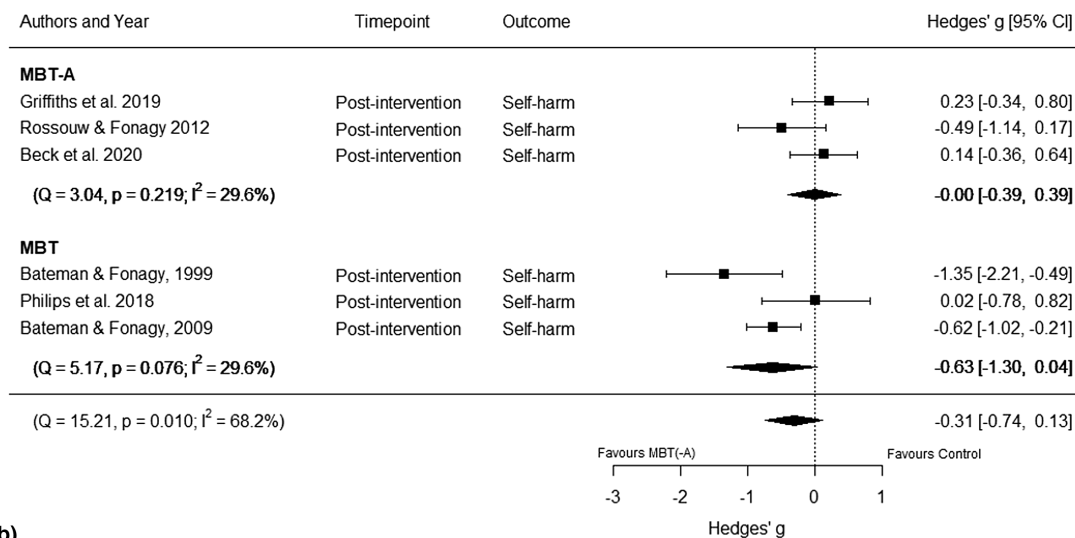
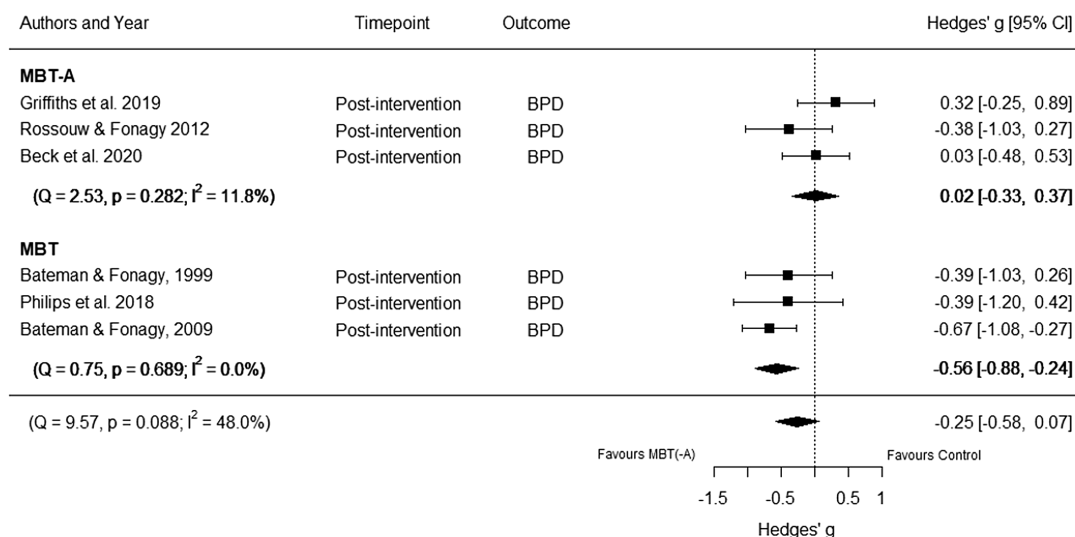


FIGURE 2 Forest plots of pre-post comparisons of the effect sizes of MBT(-A) on (a) self-harm, (b) borderline personality disorder (BPD) symptoms, and (c) depression symptoms.

(a)



(b)



(c)

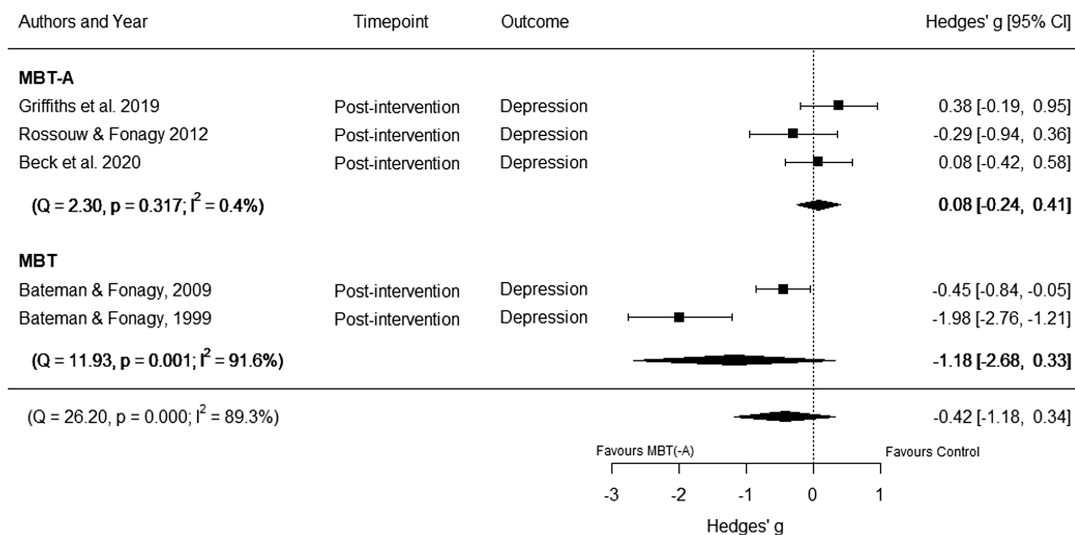


FIGURE 3 Forest plots of comparisons of the effect sizes of MBT(-A) and control interventions on (a) self-harm, (b) borderline personality disorder (BPD) symptoms, and (c) depression symptoms.

MBT-A ($I^2=11.8\%$) and MBT ($I^2=0.0\%$) studies. Overall, MBT was superior, and MBT-A was not superior to control interventions in reducing BPD scores.

Regarding the overall efficacy of MBT(-A) (combined adolescents/adults) compared to control interventions in improving symptoms of depression (see Figure 3c), we computed five effect sizes covering 288 participants (MBT-A: $n=66$, MBT: $n=71$, Control: $n=151$, $k=5$). The meta-analysis yielded no significant difference in depression scores between groups. Heterogeneity between all studies was high ($I^2=89.3\%$). Subgroup analysis of MBT-A (adolescents only) and MBT (adults only) studies also revealed no significant effect between groups on depression. Heterogeneity was very low between MBT-A studies ($I^2=0.4\%$) and very high between MBT studies ($I^2=91.6\%$). Overall, MBT(-A) was not superior to control interventions in reducing depression scores.

Meta-regressions

Multiple meta-regressions were conducted to assess whether study characteristics (treatment duration in months, dropout rates in the MBT(-A) group and mean age in years) moderate effect sizes of the pre-post comparisons (see Appendix S1A). Only treatment duration was negatively associated with effect sizes for self-harm ($b=-0.07$, $p=0.002$), such that the longer the MBT(-A) treatment, the larger the reduction in self-harm from pre to post. No other moderating effects of treatment duration, dropout rates, or mean age were found on self-harm, BPD, or depression symptoms.

We repeated these meta-regressions on the meta-analytic effects between MBT(-A) and control interventions, including RCTs only (see Appendix S1B). All three moderators were associated with effect size changes in self-harm, but not depression and BPD symptoms. Treatment duration was negatively associated with the change in effect sizes for self-harm ($b=-0.22$, $p=0.005$). That is, the longer the MBT(-A) treatment, the greater the reduction in self-harm compared to control interventions. In contrast, mean age ($b=0.10$, $p=0.045$) and dropouts ($b=0.06$, $p=0.015$) were positively associated with the change in effect sizes for self-harm. A higher sample mean age and higher dropout rates led to smaller reductions in self-harm compared to control interventions.

Exploratory sensitivity analyses

To assess a potential bias, we excluded the study by Griffiths et al. (2019) that conducted a pilot RCT of only

a brief mentalization-based introductory intervention in groups (MBT-Ai) in combination with TAU. As MBT-Ai was limited to 12 weeks of psychoeducation and experiential exercises, but did not include individual/group therapy, it cannot be equated to a full MBT-A program. A reanalysis of our outcomes (reduction of self-harm, BPD, and depression symptom severity) including only full MBT-A programs did not change our results, irrespective of control comparison (self-harm symptom severity: pre-post MBT-A trials, $k=4$, $n=171$, $g=-0.67$, 95% CI -1.32 to -0.03 , $p<0.05$; MBT-A vs. control, $k=2$, $n=103$, $g=-0.14$, 95% CI -0.75 to 0.48 , $p>0.05$; BPD symptom severity: pre-post MBT-A trials, $k=3$, $n=69$, $g=-1.11$, 95% CI -1.53 to -0.69 , $p<0.05$; MBT-A vs. control, $k=2$, $n=103$, $g=-0.13$, 95% CI -0.52 to 0.27 , $p>0.05$; depression symptom severity: pre-post MBT-A trials, $k=3$, $n=69$, $g=-1.06$, 95% CI -1.47 to -0.64 , $p<0.05$; MBT-A vs. control, $k=2$, $n=103$, $g=-0.06$, 95% CI -0.45 to 0.34 , $p>0.05$). Merely the overall effect of MBT(-A) on BPD symptom severity post-intervention became significant excluding participants from Griffiths et al. (2019) (BPD symptom severity: MBT(-A) vs. control, $k=5$, $n=264$, $g=-0.38$, 95% CI -0.68 to -0.08 , $p<0.05$).

We further conducted an exploratory subgroup analysis, to differentiate between MBT-A treatment settings (individual vs. group), again excluding Griffiths et al. (2019). The subgroup analysis of pre-post comparisons regarding self-harm revealed that MBT-A conducted in an individual setting resulted in a distinctly larger effect (self-harm symptom severity: pre-post MBT-A (individual), $k=1$, $n=20$, $g=-1.79$, 95% CI -2.78 to -0.80 , $p<0.05$) than the combined effect of MBT-A in a group setting (self-harm symptom severity: pre-post MBT-A (group), $k=3$, $n=151$, $g=-0.43$, 95% CI -0.85 to -0.02 , $p<0.05$). The difference in pre-post effects between individual and group MBT-A settings was less distinct regarding BPD and depression symptom severity (BPD symptom severity: pre-post MBT-A (individual), $k=1$, $n=20$, $g=-0.93$, 95% CI -1.66 to -0.20 , $p<0.05$; pre-post MBT-A (group), $k=2$, $n=49$, $g=-1.20$, 95% CI -1.73 to -0.68 , $p<0.05$; depression symptom severity: pre-post MBT-A (individual), $k=1$, $n=20$, $g=-1.20$, 95% CI -2.01 to -0.40 , $p<0.05$; pre-post MBT-A (group), $k=2$, $n=49$, $g=-1.01$, 95% CI -1.49 to -0.52 , $p<0.05$). It should be noted that the effect of MBT-A in an individual setting is based solely on one study covering $n=20$ adolescents in the MBT-A group (Rossouw & Fonagy, 2012). Since there was only one RCT in each subgroup (individual vs. group), no subgroup analysis was conducted on our outcomes for comparisons to control interventions.

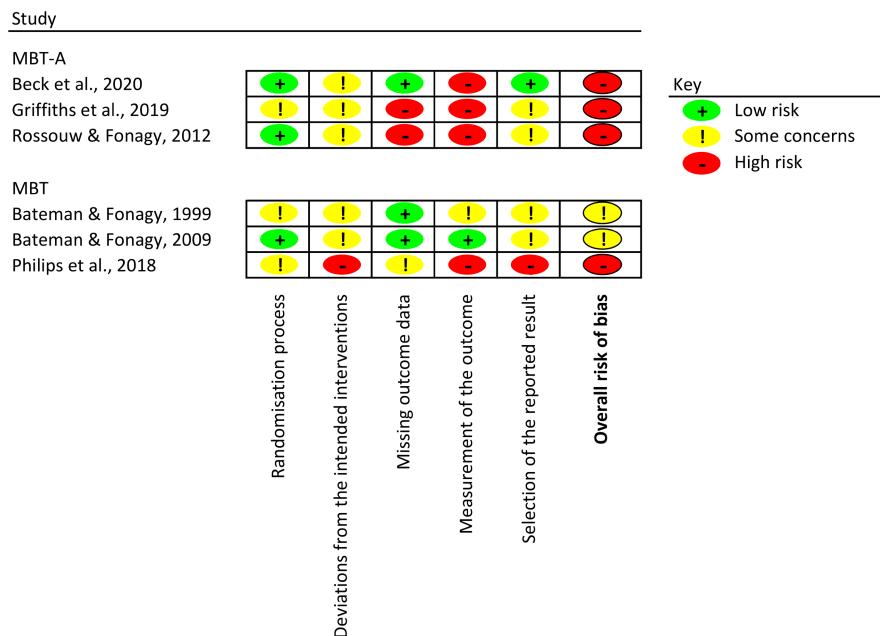


FIGURE 4 Risk of Bias Assessment.

Risk of bias

Four (66.7%) of the six included RCTs had a high risk of bias, and two (33.3%) showed a moderate risk of bias. Disagreements between the authors were present in 20.45% of all “signaling questions.” The bias risk was dominant in MBT-A RCTs, especially in the domain “measurement of the outcome.” For a more detailed overview of the detected biases by domain, see Figure 4.

DISCUSSION

Given the clinical significance and high prevalence of self-harm, especially in patients with BPD, establishing effective treatments in adolescents and adults is crucial. This meta-analysis is the first study to synthesize findings from both pre-post evaluation studies and RCTs of MBT(-A) interventions in treating self-harm and further BPD symptoms and depression in the age groups of adolescents and adults. In total, our meta-analysis included 14 studies, comprising 278 adolescents and 334 adults.

MBT(-A) demonstrated effectiveness in reducing self-harm, BPD symptoms and depression in adolescents and adults, with rather large treatment effects (ranging from $g = -1.1$ to $g = -0.82$). These findings are in line with previous systematic reviews (Malda-Castillo et al., 2019; Vogt & Norman, 2019), providing preliminary evidence of MBT(-A)'s effectiveness across studies of different methodological rigor. Although these findings seem promising, MBT(-A) did not prove to be a more efficacious treatment than control interventions, questioning its implication for

clinical practice. The only exception was MBT with superior, however, moderate efficacy ($g = -0.56$) on BPD symptoms in adults.

The main objective of our meta-analysis was to evaluate the effectiveness and efficacy of MBT(-A) for the treatment of self-harm. In this regard, MBT(-A) does not appear to add benefit compared to control interventions when focusing on self-harm. This is surprising, since improvements in mentalizing were found to be mediating factors in the reduction of self-harm in adolescents (Rossouw & Fonagy, 2012) and were further found to be the only independent predictor of suicide risk in adult psychiatric inpatients (Pompili et al., 2017).

At one level, it has been proposed that mentalization is a process within the therapeutic alliance that may be common to many forms of treatment (Fonagy & Allison, 2014). Mentalization in the context of therapy helps patients to perceive their mental states represented by the therapist, which strengthens their interpersonal understanding and challenges them to question their relationship to their environment (Brent, 2015; Fonagy & Allison, 2014; Fonagy & Bateman, 2007). Mentalization has been reasoned to be a key mechanism of change in therapeutic interventions (Brent, 2015; Fonagy & Allison, 2014). If mentalizing in the patient-therapist relationship is not an unique attribute of MBT(-A) but could also be improved in contacts within control interventions, it might explain why MBT(-A) appears no more efficacious than control conditions.

At another level, it might be possible that more in-depth training in mentalizing abilities leads to a reduction of self-harm, but that the applied group setting of MBT(-A)

in patients with BPD limits its efficacy. The group settings serve as a framework for practicing mentalizing abilities in interaction with others. In the individual setting, experiences from the group and external life are discussed in more detail (Karterud, 2015). Interestingly, three studies reviewed (Beck et al., 2020; Bo et al., 2017; Tromp et al., 2021) applied MBT-A only in group settings. The use of group settings in patients with BPD has been widely discussed in the literature (Bo et al., 2021; Karterud, 2015; McLaughlin et al., 2019). BPD symptoms encompass emotional instability and extreme beliefs within interpersonal relationships (Bo et al., 2021; Fertuck et al., 2018; Karterud, 2015). In group settings, this might allow for the possibility of emotional contagion from other group participants and therapists might not have the capacity to consistently co-regulate the individual patient's affect (Beck et al., 2020). This may cause patients to experience increased emotional arousal and dysfunctional exchanges with other group participants (Beck et al., 2020; Karterud, 2015), which has been associated with a rapid decline in mentalizing capacity (Rossouw, 2015; Smeets et al., 2009). Adolescent patients may be sensitive to the social dynamics of group interventions due to puberty-related impairments in affect regulation and an increased risk of co-influence from peers (Beck et al., 2020; Bo et al., 2021). For instance, prior studies showed that adolescents are more susceptible to being influenced by peers to engage in risky behavior (e.g., violent behavior, substance abuse) compared to adults (Gardner & Steinberg, 2005; Prinstein, 2001). Given the unique needs of adolescent patients with BPD, the improvement of mentalization might be limited in group treatments as a stand-alone therapeutic intervention. Preliminary findings from our exploratory sensitivity analysis provide first insights that MBT-A in a group format might be less effective than in an individual format in reducing self-harm, although this comparison is based on solely one MBT-A study in an individual setting comprising $n = 20$ adolescents and should therefore be interpreted with caution. Future studies should investigate if supplementing or even substituting group treatments with individual or family settings may target mentalization more effectively.

Finally, it can be questioned whether the absence of a direct and specific focus on self-harm in MBT(-A) might contribute to its limited efficacy in self-harm reduction. MBT is distinguished by its indirect approach to reduce self-harm, prioritizing the development of mentalizing skills (Barnicot & Crawford, 2019; Bateman & Fonagy, 2006; Taubner et al., 2019). The rationale is grounded in the observation that individuals, particularly during episodes of interpersonal stress, may encounter difficulties to introspectively consider their mental states and are instead overwhelmed by negative thoughts and feelings about themselves (Rossouw

& Fonagy, 2012). Self-harm represents a way to escape this negative self-evaluation (Rossouw & Fonagy, 2012). Additionally, impaired mentalization may result in social isolation, prompting individuals to use self-harm as a way to reconnect with their environment (Rossouw & Fonagy, 2012). While MBT(-A) targets underlying mentalizing deficits, it might neglect the additional acquisition of specific coping strategies (e.g., directed emotion regulation skills), which has been shown to act as a mediator in reducing self-harm in (DBT) (Barnicot et al., 2016; Neacsiu et al., 2010).

In our meta-analysis, a greater improvement in BPD symptoms over control interventions was only found in adults. It has been proposed that interventions with high mentalization demands might not be as effective in adolescents in general, due to the age-specific neurobiological changes. During puberty, adolescents exhibit a “dip” in social cognition because of the structural and functional fine-tuning of the “social brain,” a network of brain areas involved in recognizing others and evaluating their mental states (Blakemore, 2008; Bo et al., 2021; Choudhury et al., 2006; Harenski et al., 2012). Maladaptive mentalizing strategies, commonly observed in adolescence, might reflect ongoing socioemotional learning (Poznyak et al., 2019). This assumption is further supported by functional neuroimaging studies, indicating that neural activations during mentalizing differ between adolescents and adults. For instance, Burnett et al. (2009) found that adults exhibit an increased left temporal lobe activity compared to adolescents when processing social emotions that require mentalization. Blakemore et al. (2007) demonstrated that adolescents exhibit increased activation of the medial prefrontal cortex compared to adults when answering questions about intentional causality. Overall, these differences may impact the degree to which MBT can be applied for adolescents and adults in a similar manner.

Across all studies in our meta-analysis, longer treatment duration was associated with greater reductions in self-harm, while no association with BPD or depression symptoms was found. It should be noted that our included studies vary greatly in treatment duration. Most MBT-A programs were offered for 12 months (Beck et al., 2020; Bo et al., 2017; Rossouw & Fonagy, 2012), while Griffiths et al. (2019) conducted a pilot RCT of only a brief mentalization-based introductory intervention in groups (MBT-Ai) in combination with TAU for 12 weeks only. In contrast, the majority of studies offered MBT for adults for at least 18 months (Bales et al., 2012; Bateman & Fonagy, 1999, 2009; Löf et al., 2018; Philips et al., 2018), with Barnicot and Crawford (2019) offering a 12-month treatment and Kvarstein et al. (2015) even offering a 36-month treatment. Due to the heterogeneity in the reviewed studies, it is not clear, which MBT(-A) duration, intensity, and treatment components are most beneficial.

For this reason, we strongly recommend that future study designs be standardized, allowing generalizable conclusions to be drawn.

Limitations

This meta-analysis has several limitations. First, there was partially high heterogeneity in the included studies. This heterogeneity is not surprising since treatment designs (e.g., treatment durations, patient accommodation, treatment settings, treatment format), as well as the operationalization of primary and secondary outcomes, differed substantially across studies (see Table 1). Of particular note, the MBT-A studies varied greatly in terms of treatment format (group, individual, introductory group), questioning whether these can be summarized under the umbrella term MBT-A. While an introductory group limited to 12 weeks of psychoeducation and experiential exercises (see Griffiths et al., 2019) cannot be equated to a full MBT-A program, MBT-G was developed as a cost-effective alternative to the multicomponent format and serves as a stand-alone treatment with a typical duration of 12–36 months. Nevertheless, it features a lower intensity and duration than a multicomponent MBT-A. For this reason, we conducted a sensitivity analysis, which revealed no substantial changes in our results. Second, an inclusion criterion was the assessment of the primary outcome (self-harm) and studies assessing only the secondary outcomes (BPD, depression) were excluded (e.g., Jørgensen et al., 2013; Laurensen et al., 2018; Robinson et al., 2016). Therefore, the present synthesized effect of MBT(-A) on the secondary outcomes does not reflect the currently available data in the literature. Third, the analysis did not evaluate mentalizing or reflective functioning measures (Fonagy et al., 2016). Given that the overarching aim of MBT(-A) is to improve mentalizing capacity, no conclusion can be drawn whether this aim is actually achieved. Fourth, control conditions in the studies reviewed were not well described and partially included principles of specific therapeutic interventions (such as DBT, see Griffiths et al., 2019). Critically, TAU has the advantage over MBT(-A) of offering more flexibility to meet patients' needs (Witt et al., 2021). Future studies should standardize the operationalization of control conditions to isolate the actual effect of MBT(-A) and better inform research and clinical practice. Fifth, many of the studies reviewed did not differentiate between self-harm with and without suicidal intent. However, both behaviors have different underlying purposes and do not share the same etiologies and risk factors (Johnstone et al., 2022; Muehlenkamp, 2005). To better understand the relationship between mentalizing and self-harm, future studies

should consider a more differentiated operationalization. A final limitation includes the mostly low methodological quality of the included studies, and critically, all studies with MBT-A had an overall high risk of bias.

CONCLUSIONS

Despite these limitations, this meta-analysis provides the most comprehensive synthesis to date of the available evidence for the effectiveness and efficacy of MBT(-A) in reducing self-harm as well as depression and BPD symptoms among adolescents and adults. In pre-post evaluations, MBT(-A) led to improvements across all three outcomes. The analysis including only RCTs showed no superiority to active controls regarding self-harm and depression, so that MBT(-A) cannot be prioritized above other interventions in clinical practice. Merely in adults but not adolescents with BPD, MBT demonstrated a significant moderate treatment effect compared to active controls. Regarding adolescents, the treatment effects of MBT-A are lower than, for example, the effects of DBT-A. Further, there is a longer treatment duration associated with MBT-A (12–18 months) compared with DBT-A treatment (4–6 months) (Kothgassner et al., 2021). These long treatment periods might particularly hinder adolescents in their social development and in their progress at school. Following this, MBT(-A) should not be recommended as a first-line treatment for self-harm and depressive symptoms irrespective of age groups. However, future research about the effects of MBT on BPD symptoms in adults is encouraged.

CONFLICT OF INTEREST STATEMENT

None of the authors declare any conflict of interest with regard to this article. PLP received research funding from the German Federal Ministry of Education and Research (BMBF), the German Federal agency of drugs and medical devices (BfARM), the Volkswagen Foundation, the Baden-Wuerttemberg state foundation, Vienna's Landeszielsteuerungskommission, the Austria National fund, the Austrian Future Fund, and the Austrian University jubilee fund. He acted as speaker for GSK, Meducately, InfectoPharm, Janssen, and Procter & Gamble. He is an advisor for Boehringer Ingelheim and receives royalties from Hogrefe publishers. ODK received research funding from the Austrian Science Fund (FWF) and the EWSA Fond.

FUNDING INFORMATION

This project was funded by the Austrian Science Fund (FWF): KLI963-B. Open access funding is provided by Medical University of Vienna.

DATA AVAILABILITY STATEMENT

Data supporting this study are openly available in a repository of the Open Science Framework at <https://doi.org/10.17605/OSF.IO/BQMYZ>.

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How to cite this article: Hajek Gross, C., Oehlke, S.-M., Prillinger, K., Goreis, A., Plener, P. L., & Kothgassner, O. D. (2024). Efficacy of mentalization-based therapy in treating self-harm: A systematic review and meta-analysis. *Suicide and Life-Threatening Behavior*, 54, 317–337. <https://doi.org/10.1111/sltb.13044>